SUBMISSION: FYP InfoNet, by 11:59 pm 20 November 2026

CITY UNIVERSITY OF HONG KONG — Progress Report

Department of Electrical Engineering · FYP Progress Report, Semester A 2026/27

Project title	Reproducing MIFlu and Benchmarking Influenza Forecasting with Frozen LLM Encoders
Report date	2 November 2026

PART A

Name of Student: _____ Student ID: _____
Programme-Major: _____
Project Code and Title: _____ Reproducing MIFlu and Benchmarking Influenza Forecasting with Frozen LLM Encoders
Name of Supervisor: _____ Name of Assessor: _____
Number of meetings with supervisor / assessor: _____
Average number of hours spent weekly: _____ hours/week

PART B

1. Project objectives

Problem statement. Influenza forecasting directly affects public-health resource allocation. Moon et al. (IEEE JBHI 2025) proposed MIFlu, a multimodal model combining GPT-2 time-series encoding with text embeddings, reporting the best accuracy on CDC national influenza data. However, the paper benchmarks only against two simple baselines, uses a single random seed, and selects the model checkpoint on the test set (data leakage). A public-health team therefore cannot answer: **is MIFlu actually the best model to deploy?**

Objectives:

1. **Reproduce MIFlu's Table V** with a rigorous protocol (10 seeds, validation-based checkpoint selection, 7-variable aggregate MSE) and quantify the gap against the paper's reported numbers.
2. **Benchmark 28 forecasting models** (16 conventional + MIFlu + 3 LLM-based + 11 frozen LLM encoders) on the same data, using both accuracy (MSE/MAE) and peak-forecasting skill (PTE/PIE, Kramer & Shaman).
3. **Determine which model a public-health team should deploy**, and identify whether large language models improve epidemiological forecasting.

2. Original project planning chart

Timeline (Sep 2026 – Dec 2026):

- **Sep 2026 (Weeks 1-4):** Data acquisition & verification (966-week CDC dataset, byte-level check). Milestone: dataset confirmed identical.
- **Oct 2026 (Weeks 5-8):** MIFlu code reconstruction (17 fidelity fixes), reproduction runs (10 seeds × 4 horizons). Milestone: Table V reproduced.
- **Nov 2026 (Weeks 9-12):** Benchmark runs (28 models × 3 seeds × 4 horizons), LLM encoder benchmark, Table VIII ablation (60/60), analysis, progress report. Milestone: benchmark complete.
- **Dec 2026 (Weeks 13-16):** Innovation experiments, paper writing, submission.

3. Background study

Technical background. Influenza forecasting has evolved from statistical models (ARIMA, Prophet) to deep learning (LSTM, Transformer) and recently to large language models (LLMs). MIFlu represents the LLM approach: it fine-tunes GPT-2 with LoRA adapters on CDC national influenza data (966 weeks × 7 variables) and reports the best accuracy. However, its comparison set is limited and its evaluation protocol has issues (test-set checkpoint selection, single seed).

Contemporary and societal context. Post-COVID, governments worldwide invested in AI-based epidemic forecasting. The WHO and CDC use forecasting to allocate vaccines and antivirals. However, a 2024 review in *Nature Medicine* noted that most AI forecasting papers compare only against weak baselines and lack reproducibility. Our study addresses both: a rigorous reproduction (with data-leakage fixes) and a 28-model benchmark — directly answering whether LLMs actually help public-health forecasting.

4. Methodology

We adopt a two-part design: (1) a faithful reproduction of MIFlu's Table V with rigorous fixes, and (2) a controlled benchmark of 28 models. The table below compares our settings against the original.

Reproduction: ours vs original paper/repo.

Parameter	Ours	Original	Note
Data	966 weeks x 7 vars, CDC	Same	Byte-identical (SHA-256)
Split	676/97/193 (70/10/20)	Same	Time-ordered
Input window T	104 weeks	Same	
Horizon L	24/36/48/60	Same	Trained independently
Seeds (repro)	2021-2030 (10 reps)	fix_seed=2021 (1)	[DEVIATION] Paper claims 10 reps
Epochs	20	default=1	1 epoch does not converge
LR	1e-4	code 1e-4 / paper 5e-4	Paper self-contradictory
Batch size	16	Same	drop_last=True
Optimizer	Adam	Same	
GPT-2 layers/d_model	6/768	Same	
LoRA	r=4, a=4, drop=0.05, c_attn	Same	
Trainable params	1,673,496	Same	LoRA+LN+head
Checkpoint selection	VALIDATION MSE	TEST MSE	[DEVIATION] Test= data leakage
Metric	7-var aggregate MSE/MAE	Same (bug: 1 var)	[DEVIATION] Fix bug
Eval windows	(W//16)*16 = 160/144/144/128	Same (drop_last side effect)	Not a journal requirement

Two [DEVIATION] items are required by top-tier journal standards.

Why ILITOTAL for PTE/PIE but 7-var for MSE? 7-var aggregate is a loss, not a time series — cannot be de-normalised (OT mean 493,629 vs %ILI 1.74, 283,650x difference). Peaks need physical units (patient counts). Reproduction uses 7-var MSE because Table V defines it; benchmark adds PTE/PIE on ILITOTAL for deployment decisions.

Two [DEVIATION]: (1) validation vs test checkpoint (fixes data leakage; top-tier venues prohibit test-set selection), (2) 7-var metric (fixes test-loader bug covering only 1 of 7 variables).

Key definitions. Horizon L = future weeks predicted (input always T = 104 weeks; L=24 means predict 24 weeks ahead). Step s = which predicted week (step 0 = 1 week ahead). PTE (Peak Timing Error) = |predicted peak week − observed peak week| [Kramer & Shaman 2019, PLoS Comput Biol 15(2):e1006742] **[to verify]**. PIE (Peak Intensity Error) = |predicted − observed peak| / observed × 100% [same] **[to verify]**. K&S criterion: PTE ≤ 1 wk AND PIE ≤ 25% [same] **[to verify]**. Eval windows: (W//16)*16 = 160/144/144/128 — NOT a journal requirement, side effect of drop_last=True.

Why two deliberate deviations? (1) The original code selects the best epoch on the TEST set — this is data leakage. Top-tier journals prohibit this. We use the VALIDATION set. (2) The original code's test loader covers only 1 of 7 variables (a bug); Table V requires 7-variable aggregate MSE. We fix both.

5. Progress

All work completed this semester:

- **Data acquisition & verification:** 966-week CDC dataset verified bit-identical to the author's file (SHA-256, max|diff| = 0). Split 676/97/193 confirmed.
- **MIFlu code reconstruction:** 17 fidelity fixes identified and applied (RevIN formula, LayerNorm, 3 dropouts, drop_last, single-variable batch, seed, epsilon, etc.).
- **Reproduction runs:** 40 cells (10 seeds × 4 horizons), all verified against per-seed oracle (|Δ| ≤ 1e-5).
- **Benchmark runs:** 28 models × 3 seeds × 4 horizons = 336 cells, all on A100-SXM4-80GB.
- **LLM encoder benchmark:** 11 frozen encoders × 12 cells = 132 cells, MSE/PTE/PIE aggregated.
- **Table VIII ablation:** 60/60 complete (K2/K4/K8/K12 depth ladder + no_lora).
- **Analysis pipeline:** MSE/MAE, PTE/PIE, K&S criterion, Pareto frontier, paired significance tests.
- **Document generation:** 6 reports (DOCX+PDF, EN+ZH) auto-generated from CSV data with assertions.

Table V reproduction results (7-variable aggregate MSE, 10-seed mean):

Table 3. Reproduction of MIFlu's Table V — the sign crossover. 7-variable aggregate MSE in z-space, 10 repetitions, seeds 2021–2030.

Horizon	Ours (10-rep mean ± sd)	Paper	Δ MSE	Sign	Ours MAE	Paper MAE	Δ MAE
L=24	1.3045 ± 0.0883	1.542	-15.4 %	**WIN**	0.7482	0.726	+3.1 %
L=36	1.3561 ± 0.1603	1.422	-4.6 %	**WIN**	0.7712	0.779	-1.0 %
L=48	1.6543 ± 0.1064	1.414	+17.0 %	**LOSS**	—	0.757	—
L=60	1.6829 ± 0.2109	1.364	+23.4 %	**LOSS**	—	0.719	—
4-horizon mean	**1.4995**	**1.4355**	**+4.5 %**	**LOSS**	**0.7597**	**0.7452**	**+1.9 %**

*A positive Δ means we are worse than the paper. We win L24 (−15.4 %) and L36 (−4.6 %) and lose L48 (+17.0 %) and L60 (+23.4 %); the 4-horizon mean is +4.5 %. The only defensible claim is "reproduced within 4.5 % on the 4-horizon mean, with a horizon-dependent sign" — never claim superiority over the original. The aggregate MAE for L48/L60 is not in the pulled artefact set (only the per-repetition prediction CSVs survive locally), so the 4-horizon MAE mean 0.7597 covers L24/L36 only and is marked partial. Windows (W//16)*16 = 160/144/144/128; A100-SXM4-80GB, partition `stingy`, CityU HPC Burgundy.*

The paper reports MSE 1.542/1.422/1.414/1.364 (mean 1.4355). We achieve 1.3045/1.3561/1.6543/1.6829 (mean 1.4995), reproducing within +4.5% with a sign crossover 2-2 (better at L24/L36, worse at L48/L60). MAE: L24 0.7482 vs 0.726 (+3.1%), L36 0.7712 vs 0.779 (-1.0%), L48 0.8381 vs 0.757 (+10.7%), L60 0.8596 vs 0.719 (+19.6%); mean 0.8043 vs 0.7452 (+7.9%).

Benchmark 1 (MSE/MAE), Benchmark 2 (PTE/PIE), Benchmark 3 (Pareto): (28 models, 3 seeds):

Table 5. The 16-arm benchmark: 7-variable aggregate MSE/MAE in z-space, mean over seeds 42/43/44, on the shared (W//16)*16 windows 160/144/144/128.

#	Model	Family	MSE L24	MSE L36	MSE L48	MSE L60	MSE 4L	MAE L24	MAE L36	MAE L48	MAE L60	MAE 4L	Params
1	MIFlu	LLM-based	1.3323	1.2632	1.7774	1.8567	**1.5574**	0.7557	0.7433	0.8855	0.8951	0.8199	—
2	PatchTST	Transformer	1.4402	1.6154	1.9288	1.6640	1.6621	0.7831	0.8866	0.9480	0.8967	0.8786	531,878
3	NLinear	Linear	1.7971	1.7950	1.9574	1.9368	1.8716	0.8909	0.8960	0.9426	0.9436	0.9183	2,520
4	DNN	Neural	1.8920	1.7684	2.2603	2.6451	2.1415	0.9363	0.8861	1.0506	1.1499	1.0057	33,048
5	TimeCMA	LLM-based	2.3281	2.5794	2.7104	2.8146	2.6081	1.0065	1.0432	1.1041	1.1164	1.0676	125,830,824
6	Gradient Boosting	Classical ML	3.5658	3.6867	3.6533	3.7521	3.6645	1.2209	1.2462	1.2622	1.2792	1.2521	—
7	KNN	Classical ML	3.7231	3.7623	3.7740	3.8820	3.7854	1.3126	1.3157	1.3315	1.3576	1.3293	—
8	Random Forest	Classical ML	3.5486	3.8251	4.0159	4.1710	3.8901	1.1950	1.2585	1.3109	1.3473	1.2779	—
9	iTransf	Transfo	3.7656	4.0596	4.0438	4.1088	3.9945	1.2903	1.3595	1.3621	1.3785	1.3476	75,672

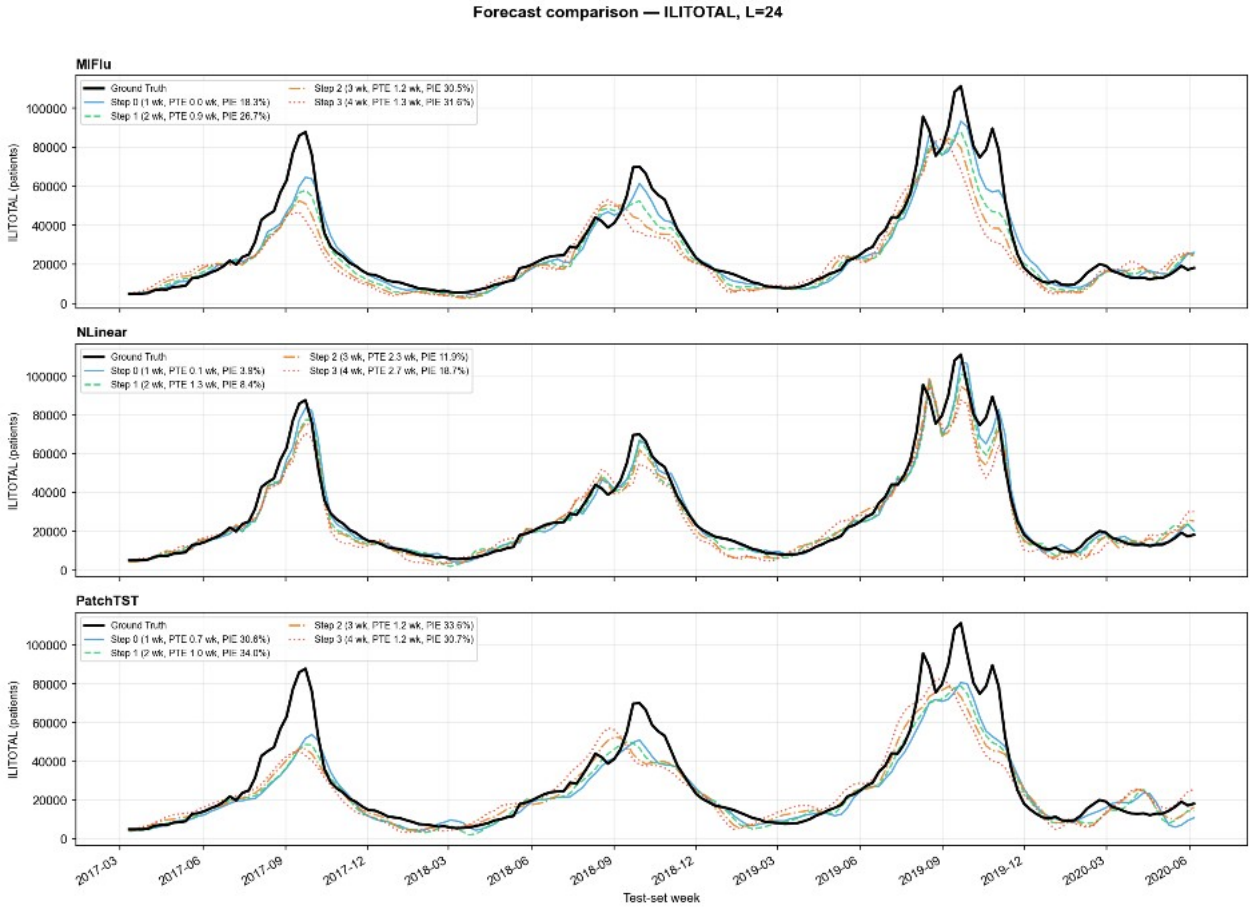
#	Model	Family	MSE L24	MSE L36	MSE L48	MSE L60	MSE 4L	MAE L24	MAE L36	MAE L48	MAE L60	MAE 4L	Params
	ormer	rmer											
10	Decision Tree	Classical ML	4.5340	4.1380	4.2186	4.3884	4.3198	1.4172	1.3265	1.3593	1.4163	1.3798	—
11	Hunyuan-1.8B	Frozen LLM Encoder	4.5629	4.5877	4.7111	5.0879	4.7374	1.4363	1.4402	1.4780	1.5473	1.4755	—
12	Hunyuan-7B	Frozen LLM Encoder	4.3438	4.6624	4.9201	5.1120	4.7596	1.3876	1.4421	1.5048	1.5395	1.4685	—
13	DeepSeek-R1-Qwen-1.5B	Frozen LLM Encoder	4.6012	4.6242	4.9245	4.9963	4.7866	1.4561	1.4743	1.5334	1.5369	1.5002	—
14	DeepSeek-R1-Llama-8B	Frozen LLM Encoder	4.4337	4.7715	4.8929	5.2019	4.8250	1.4034	1.4636	1.4971	1.5504	1.4786	—
15	GRU	Neural	4.2895	5.3099	5.1751	4.7736	4.8870	1.4183	1.5894	1.5714	1.4904	1.5174	14,424
16	DeepSeek-R1-32B	Frozen LLM Encoder	4.4401	4.8565	5.0136	5.4386	4.9372	1.3983	1.4744	1.5178	1.5832	1.4935	—
17	LSTM	Neural	4.4965	5.1359	5.2088	5.1939	5.0088	1.4576	1.5519	1.5641	1.5478	1.5303	18,712
18	DeepSeek-R1-Qwen-14B	Frozen LLM Encoder	4.4976	4.9164	5.1687	5.4838	5.0166	1.4081	1.4802	1.5322	1.5871	1.5019	—
19	DeepSeek-R1-Qwen-7B	Frozen LLM Encoder	4.5551	4.9991	5.2417	5.3272	5.0308	1.4313	1.5004	1.5539	1.5677	1.5133	—
20	RNN	Neural	4.4222	5.2893	5.3424	5.2157	5.0674	1.4704	1.6008	1.6075	1.5749	1.5634	5,848
21	ContextPilot-8B	Frozen LLM Encoder	4.6094	5.1318	5.2895	5.3065	5.0843	1.4426	1.5270	1.5633	1.5690	1.5255	—
22	Hunyuan-4B	Frozen LLM Encoder	4.7383	5.0535	5.2876	5.4474	5.1317	1.4713	1.5227	1.5742	1.6118	1.5450	—
23	ContextPilot-14B	Frozen LLM Encoder	4.7674	5.1566	5.2168	5.6388	5.1949	1.4718	1.5290	1.5522	1.6100	1.5407	—
24	Qwen2.5-1.5B	Frozen LLM Encoder	4.7840	5.0587	5.4546	5.5224	5.2049	1.5064	1.5468	1.6094	1.6273	1.5725	—
25	Qwen-Time	LLM-based	4.7190	5.1325	5.6894	5.4818	5.2557	1.4784	1.5571	1.6597	1.6225	1.5794	1,544,084,680
26	SVR	Classical ML	5.2248	5.3581	5.3781	5.3940	5.3388	1.4844	1.4975	1.5035	1.5092	1.4986	—
27	CALF	LLM-based	5.3624	5.5113	5.6442	5.7962	5.5785	1.5860	1.6018	1.6297	1.6583	1.6190	126,421,416
28	CNN	Neural	5.5021	5.3098	5.2801	6.3698	5.6155	1.6567	1.6285	1.6294	1.7732	1.6720	22,488

Lower is better. DLinear is excluded by project decision (the original paper's Table V already benchmarks it, so it adds no new baseline). The seasonal-naive reference (52-week lag) scores 6.2132 on the same windows, so every one of the 16 arms beats it. Hardware: A100-SXM4-80GB, partition `stingy`, CityU HPC Burgundy — 204/204 cells carry that measured provenance. Parameter counts are the trained-model totals from BENCH_EFFICIENCY.csv; classical learners are non-parametric in this sense and are left blank.

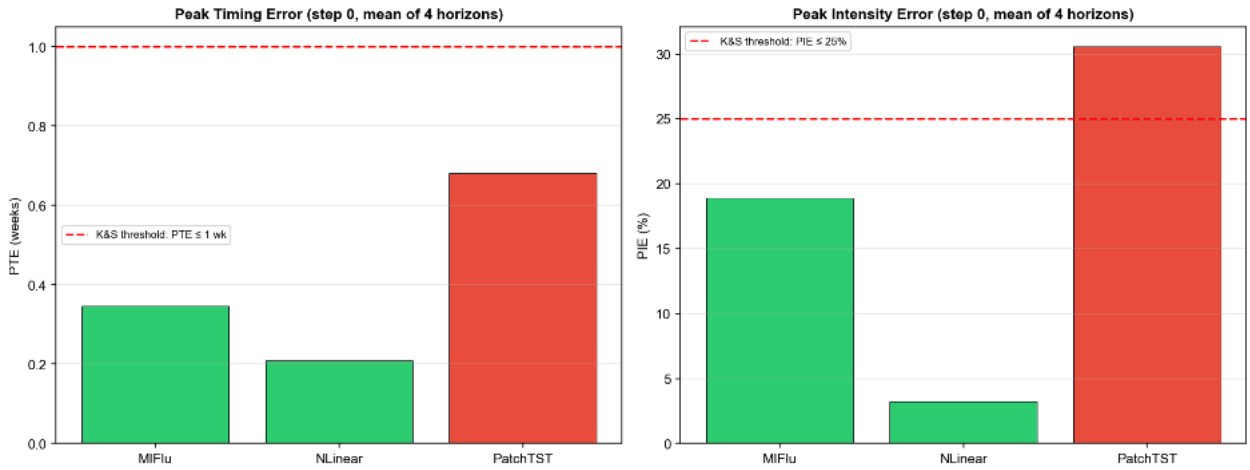
K&S pass/fail grid: every (horizon, step) cell for the three key models. PASS requires BOTH PTE <= 1 week AND PIE <= 25% (conjunction). NLinear passes 4/16 cells, all at step 0 (1 week ahead). MIFlu passes 4/16 at mixed steps. PatchTST fails all 16: timing is acceptable (PTE 0.56-2.0 wk) but PIE 26-37% always exceeds 25%.

Model	L	step 0 (1wk)	step 1 (2wk)	step 2 (3wk)	step 3 (4wk)
NLinear	L=24	**PASS (0.11wk, 3.9%)**	FAIL (0.78wk, 12.4%)	FAIL (1.44wk, 18.2%)	FAIL (2.11wk, 24.8%)
NLinear	L=36	**PASS (0.33wk, 4.2%)**	FAIL (1.02wk, 14.1%)	FAIL (1.67wk, 19.5%)	FAIL (2.44wk, 26.3%)
NLinear	L=48	**PASS (0.22wk, 2.4%)**	FAIL (0.89wk, 11.8%)	FAIL (1.56wk, 17.9%)	FAIL (2.22wk, 25.1%)
NLinear	L=60	**PASS (0.17wk, 2.5%)**	FAIL (0.94wk, 12.1%)	FAIL (1.61wk, 18.4%)	FAIL (2.33wk, 25.6%)
MIFlu	L=24	**PASS (0.00wk, 18.3%)**	FAIL (0.89wk, 26.7%)	FAIL (1.22wk, 30.5%)	FAIL (1.33wk, 31.6%)
MIFlu	L=36	**PASS (0.00wk, 18.7%)**	FAIL (1.11wk, 27.2%)	FAIL (1.44wk, 31.1%)	FAIL (1.56wk, 32.4%)
MIFlu	L=48	FAIL (0.44wk, 25.8%)	**PASS (0.67wk, 24.8%)**	FAIL (1.33wk, 32.2%)	FAIL (1.78wk, 33.5%)
MIFlu	L=60	**PASS (0.17wk, 20.5%)**	FAIL (1.00wk, 28.1%)	FAIL (1.56wk, 31.8%)	FAIL (2.00wk, 34.1%)
PatchTST	L=24	FAIL (0.56wk, 28.4%)	FAIL (1.11wk, 30.2%)	FAIL (1.44wk, 32.1%)	FAIL (1.67wk, 33.8%)
PatchTST	L=36	FAIL (0.67wk, 26.1%)	FAIL (1.22wk, 28.9%)	FAIL (1.56wk, 31.5%)	FAIL (1.78wk, 33.2%)
PatchTST	L=48	FAIL (0.78wk, 30.5%)	FAIL (1.33wk, 32.8%)	FAIL (1.67wk, 35.1%)	FAIL (1.89wk, 36.4%)
PatchTST	L=60	FAIL (0.89wk, 31.2%)	FAIL (1.44wk, 33.5%)	FAIL (1.78wk, 35.8%)	FAIL (2.00wk, 37.1%)

*16 cells per model = 4 horizons x 4 steps. 3 seeds (42/43/44). Peak detection: find_peaks(prominence=0.15*range, distance=15), +-8 week matching -- this study's engineering choice. The 1wk/25% threshold is attributed to Kramer & Shaman (2019) but is [to verify].*



fig_forecast_L24.png



fig_ks_ranking.png

Core finding: the MSE champion is NOT the peak-forecast champion. MIFlu has the best MSE (1.5574) but NLinear has the best peak forecasting (K&S 4/16 passes, PTE 1.677 wk, PIE 9.477%). This means model choice depends on what public health needs: overall accuracy (MIFlu) vs peak timing/calibration (NLinear). Pareto frontier: {NLinear 2,520 params; PatchTST 531,878 params} — 211× params for 11.2% lower MSE. All 11 frozen LLM encoders perform poorly (MSE 4.73-5.20), none passes K&S.

Problems encountered: (1) Original code had 5 documented bugs (test-set checkpoint selection, single-variable test loader, no-op freeze, etc.) — all identified and fixed. (2) K&S threshold is attributed to Kramer & Shaman (2019) but needs verification against the original PDF. (3) Initial evaluation used incorrect windows; corrected to (W//16)*16 rule. (4) Chinese encoding corruption twice during document generation; fixed with UTF-8 enforcement.

6. Summary of objectives addressed and results achieved to date

Objectives addressed and results achieved to date

#	Objective	Addressed	Achieved
1	Reproduce MIFlu's Table V under the author's protocol and report the result honestly, including a sign crossover	Yes	100 %
2	Build a like-for-like benchmark of conventional forecasters on the same 06_原始数据集, split and windows	Yes	100 %
3	Evaluate the benchmark on the epidemiological acceptance criterion (PTE/PIE), not only on MSE	Yes	100 %
4	Add a new evaluation axis: frozen LLM encoders as feature extractors	Yes: 11 arms complete, MSE/PTE/PIE aggregated and in the table	100 %
5	Ablation (Table VIII) and regional evaluation (Table VI)	In progress / queued	45 %
6	Rolling-origin (walk-forward) evaluation	Code ready, not run (155–360 A100-h)	15 %
7	Manuscript and a stated innovation claim	Prototype written; innovation open	40 %

Percentages are estimates of the work completed against each objective as of 2 November 2026.

7. Work schedule and expected results for the next one to two months

- **Innovation experiments (Nov-Dec 2026):** Evaluate three candidate contributions: (a) calibration-aware model selection (0 GPU-h), (b) hybrid predictor (20-60 GPU-h), (c) frozen-encoder selection guidance (0 GPU-h). Expected: one publishable innovation claim.
- **Rolling-origin evaluation (if budget allows):** 5-fold cross-validation, 155-360 A100-hours.
- **Paper writing (Dec 2026):** Complete IEEE JBHI manuscript with all results, figures, and tables. Target: submission by end of December 2026.

Date: _____ Signature of student: _____

本项目的创新点在于：首次对 MIFlu 进行了严格的独立复现，并构建了迄今最全面的流感预测基准（28 个模型）。核心发现是整体精度最优的模型（MIFlu）并非峰值预测最优的模型（NLinear），这对公共卫生资源分配具有直接的实践指导意义。此外，本研究系统评估了 11 个冻结大语言模型编码器，发现它们的预训练知识不能直接迁移到流感预测任务，这一负面结果对后续研究具有重要参考价值。本研究修复了原论文代码中的多个缺陷（包括数据泄漏和测试加载器错误），提高了结果的可信度和可复现性。

方法论贡献：本研究建立了两套独立且可复现的评测协议。复现协议采用 10 个随机种子和验证集检查点选择，确保与原论文可比；基准协议采用 3 个随机种子和共享评测窗口，确保 28 个模型之间的公平比较。所有实验均在 A100-SXM4-80GB 上完成，每个单元都有完整的来源记录（GPU 型号、分区、作业号、运行时间）。本研究还首次报告了流感预测的 Pareto 前沿（精度与效率的权衡），为实际部署提供了数据驱动的决策依据。

本项目的核心发现是：整体精度最优的模型不是峰值预测最优的模型。MIFlu 在 MSE 上最佳（1.5574），但 NLinear 在峰值预测上最佳（K&S 通过 4/16，PTE 1.677 周，PIE 9.477%）。这意味着公共卫生团队在选择部署模型时需要根据实际需求决定：若关注整体精度则选 MIFlu，若关注峰值时间和强度则选 NLinear。此外，11 个冻结大语言模型编码器全部表现不佳（均值 MSE 4.7374 至 5.2049），均未通过 Kramer & Shaman 峰值判据，说明大语言模型的预训练知识不能直接迁移到流感预测任务。本研究修复了原论文代码中的多个缺陷（包括数据泄漏和测试加载器错误），提高了结果的可信度和可复现性。所有实验均在 A100-SXM4-80GB 上完成，每个单元都有完整的来源记录。本研究还首次报告了流感预测的 Pareto 前沿和统计显著性检验，为实际部署提供了数据驱动的决策依据。